Ideation Phase
Define the Problem Statements

Date	31 January 2026
Team ID	LTVIP2026TMIDS24253
Project Name	**ELECTRICITY CONSUMPTION ANALYSIS**
Maximum Marks	2 Marks

CUSTOMER PROBLEM STATEMENT TEMPLATE

Customer Problem Statement Template:

People want to understand electricity consumption patterns across different regions and time periods, but energy data is often complex and scattered across multiple sources. Users need a simple and interactive way to visualize electricity consumption information so they can identify consumption patterns, support energy planning decisions, and make better resource allocation choices.

PROJECT PROBLEM ANALYSIS TABLES

Table 1: Problem Identification

Problem Category	**Specific Issue**	**Impact on Users**	**Urgency Level**
Data Complexity	16,599 records across 35 states	Difficult to identify patterns	High
Geographic Distribution	5 regions with different consumption patterns	Hard to compare regional performance	High
Temporal Analysis	2019-2020 monthly data	Time-consuming trend analysis	Medium
Data Visualization	Multiple data formats and sources	Inconsistent presentation	High
Decision Making	Lack of interactive t	Poor resource plan	Critical

Problem Category	Specific Issue	Impact on Users	Urgency Level
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Table 2: User Needs Assessment

User Type	Primary Need	Current Challenge	Desired Solution
Energy Planners	Regional consumption analysis	Manual data compilation	Interactive dashboard
Policy Makers	Policy impact assessment	Complex data interpretation	Visual insights
Researchers	Pattern identification	Scattered data sources	Centralized platform
General Public	Understanding consumption	Technical jargon	Simple visualizations
Students	Educational content	Academic complexity	Learning tools

Table 3: Solution Requirements

Requirement Category	Specific Need	Implementation Priority	Success Metric
Data Visualization	12 professional charts	Critical	User engagement >80%
Web Application	Interactive interface	High	Load time <3 seconds
Analysis Scenarios	Focused insights	High	Scenario completion >90%
Documentation	User guides	Medium	Documentation coverage 100%
Mobile Access	Responsive design	Medium	Mobile compatibility 95%

Table 4: Technical Challenges

Challenge Area	Specific Issue	Complexity Level	Solution Approach
Data Integration	Multiple data sources	High	Centralized database
Visualization Design	12 different chart types	Medium	Tableau Desktop
Web Development	Flask application	Medium	Professional framework
Performance	Large dataset handling	High	Optimization techniques
User Experi	Intuitive navigation	Medium	User-centered desig

Challenge Area	Specific Issue	Complexity Level	Solution Approach
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Table 5: Project Benefits

Stakeholder	Primary Benefit	Secondary Benefits	Value Proposition
Energy Sector	Better planning decisions	Cost optimization	Data-driven insights
Government	Policy development	Public transparency	Evidence-based governance
Academic	Research platform	Educational tool	Knowledge advancement
General Public	Energy awareness	Personal insights	Informed citizenship
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DETAILED PROBLEM STATEMENT

Primary Problem:

People want to understand electricity consumption patterns across different regions and time periods, but energy data is often complex and scattered across multiple sources. Users need a simple and interactive way to visualize electricity consumption information so they can identify consumption patterns, support energy planning decisions, and make better resource allocation choices.

Secondary Problems:

- Energy planners struggle with regional consumption analysis due to data fragmentation
- Policy makers lack visual tools for policy impact assessment
- Researchers face challenges in identifying consumption patterns from scattered data
- General public finds energy data too technical to understand
- Students need educational tools for learning about energy consumption

Solution Requirements:

- Interactive dashboard with 12 professional visualizations
- Web application with responsive design for all devices
- Analysis scenarios for focused insights (time patterns, seasonal variations, sector-specific)
- Complete documentation and user guides
- Mobile-friendly interface for accessibility

PROJECT IMPACT MATRIX

Table 6: Impact Assessment

Impact Area	**Current State**	**Target State**	**Improvement Factor**
Data Accessibility	Complex, scattered	Centralized, simple	10x improvement
Visualization Quality	Basic charts	Professional dashboards	5x enhancement
User Experience	Technical, difficult	Intuitive, interactive	8x better UX
Decision Support	Manual analysis	Automated insights	6x faster decisions
Educational Value	Academic complexity	Interactive learning	4x engagement

SUCCESS METRICS TABLE

Table 7: Key Performance Indicators

KPI Category	**Specific Metric**	**Target Value**	**Measurement Method**
Technical Performance	Page load time	<3 seconds	Google PageSpeed
User Engagement	Time on site	>5 minutes	Analytics tracking
Visualization Usage	Chart interactions	>80% users	User behavior analysis
Mobile Usage	Mobile visitors	>40% total	Device analytics
Documentation Access	Guide views	>1000 views	Content analytics

PROJECT ROADMAP TABLE

Table 8: Implementation Timeline

Phase	**Duration**	**Key Deliverables**	**Success Criteria**
Phase 1	2 weeks	Data analysis and cleaning	16,599 records processed
Phase 2	2 weeks	12 Tableau visualizations	All charts functional
Phase 3	2 weeks	Flask web application	15+ routes working
Phase 4	1 week	Documentation and testing	Complete guides ready
Phase 5	1 week	Demo materials and deployment	Professional presentation

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RESOURCE ALLOCATION TABLE

Table 9: Team Roles and Responsibilities

Team Member	**Primary Role**	**Key Responsibilities**	**Skills Utilized**
Member 1	Data Analyst	Data processing, Tableau visualization	Data analysis, statistics
Member 2	Web Developer	Flask application, frontend	Python, HTML/CSS/JS
Member 3	UI/UX Designer	Interface design, user experience	Design principles, Bootstrap
Member 4	Project Manager	Documentation, testing, coordination	Project management, quality assurance

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RISK MITIGATION TABLE

Table 10: Risk Assessment and Mitigation

Risk Category	**Specific Risk**	**Probability**	**Impact**	**Mitigation Strategy**
Technical	Tableau integration issues	Medium	High	Early testing, fallback options
Data	Data quality problems	Low	Medium	Data validation, cleaning
Performance	Slow loading times	Medium	Medium	Optimization techniques
User Adoption	Complex interface	Low	High	User testing, feedback
Timeline	Development delays	Medium	Medium	Buffer time, prioritization

PROJECT SUCCESS FRAMEWORK

Table 11: Success Criteria

Success Dimension	**Specific Criteria**	**Measurement Method**	**Target Achievement**
Technical Excellence	All features functional	Testing validation	100% completion
User Satisfaction	Positive user feedback	Surveys and reviews	90% satisfaction
Professional Quality	Industry standards	Expert evaluation	Professional grade
Portfolio Value	Career advancement	Job market response	Enhanced opportunities
Academic Value	Learning outcomes	Knowledge assessment	Comprehensive understanding

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🌟 FINAL PROBLEM STATEMENT SUMMARY

People want to understand electricity consumption patterns across different regions and time periods, but energy data is often complex and scattered across multiple sources. Users need a simple and interactive way to visualize electricity consumption information so they can identify consumption patterns, support energy planning decisions, and make better resource allocation choices.

This comprehensive problem statement addresses the core challenges faced by energy planners, policy makers, researchers, and the general public in understanding complex electricity consumption data, providing a clear foundation for developing a professional, interactive, and user-friendly solution.